Chat Application HLD - Master Study Guide (Blueprint Edition)

Important Note

A true textbook-quality guide covering all requested topics in depth (100+ pages with detailed explanations, diagrams, trade-offs, interview scripts and 60+ FAQs) exceeds what can be generated reliably in a single response. This document provides the complete structure and detailed chapter objectives that will be expanded in subsequent parts.

1. Foundations of Chat Systems

• What is a chat system?

• Characteristics of chat workloads

• Why chat systems are distributed

• Latency expectations

• Real-world examples

2. Requirement Gathering

• Functional requirements

• Non-functional requirements

• Clarifying questions

• Scope control

3. Capacity Estimation

• Users, QPS, storage

• Bandwidth

• Concurrent WebSocket connections

• How estimates drive architecture

4. Evolution of Architecture

• Single server

• Scaling issues

• Load balancer

• Distributed architecture

5. WebSockets Deep Dive

• HTTP vs Polling vs SSE vs WebSockets

• Handshake

• Frames

• Heartbeats

• Reconnect

• Scaling millions of sockets

6. Load Balancers

• L4 vs L7

• Sticky sessions

• Algorithms

• Failure handling

7. Chat Servers

• Connection manager

• Routing

• Authentication

• Presence

• Rate limiting

8. Redis

• Data structures

• TTL

• Presence

• Routing

• Cluster

• Sentinel

9. Kafka

• Topics

• Partitions

• Offsets

• Consumer groups

• Ordering

• Retries

• Exactly-once tradeoffs

10. Databases

• SQL vs NoSQL

• Schema

• Partition keys

• Indexes

• Replication

• Sharding

11. Presence Service

• Heartbeats

• Online

• Last seen

• Typing

12. Message Delivery

• Online

• Offline

• Acknowledgements

• Retries

13. Group Chat

• Fan-out strategies

• Large groups

14. Media Pipeline

• Uploads

• Object storage

• CDN

• Thumbnails

15. Scaling

• 1M to 1B users

• Hot partitions

• Caching

16. Multi-region

• Replication

• Geo routing

• Conflict resolution

17. Security

• TLS

• JWT

• E2EE

• Rate limiting

18. Failure Scenarios

• Redis failure

• Kafka failure

• DB failure

• Network partitions

19. Interview Walkthrough

• End-to-end answer

• Common mistakes

20. Detailed FAQs

• 60+ questions with detailed answers

Example Detailed FAQ Format

Why use WebSockets instead of HTTP polling?

Explain request-response limitations, unnecessary polling traffic, latency, server push capability, persistent TCP connection, heartbeat mechanism, scalability considerations and trade-offs.

Why Redis instead of a database for presence?

Explain transient state, TTL, in-memory performance, heartbeat updates, routing lookups and why durability is unnecessary for presence.

Why Kafka if messages are stored in Cassandra?

Explain decoupling, buffering, retries, durability, analytics consumers, search indexing, notification consumers and resilience during traffic spikes.

Roadmap

The complete handbook will expand every bullet into multiple pages with architecture diagrams, sequence diagrams, internal working, trade-offs, alternatives, interviewer expectations, implementation details, real-world examples and comprehensive FAQ answers.